



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: A1+TA1

Programme Name & Branch : B.TECH
Course Code and Course Name : BCSE305L - EMBEDDED SYSTEMS
Faculty Name(s) : ALL FACULTIES
Class Number(s) : ALL
Date of Examination : 16-03-2025
Exam Duration : 90 minutes

Maximum Marks: 50

General instruction(s):

- Answer All Questions

Q. No	Question	M
1.	Construct an FSM Model for the Given Scenario: "Autonomous Vehicle Component Classifier" Specification: - Environment: Automotive plant using robots for sorting, inspection, and packaging. - Component Storage: Bins for engine parts, tires, sensors, and electronics. - Component Types: Includes pistons, brake pads, airbags, and control units. - Robotic System: Mobile robot with a 360° arm for handling components Requirements: - Sorting Criteria: Components sorted by type, size, quality, and weight. - Collision Avoidance: Implemented to prevent damage to components, arms, or equipment. - States, Events, and Actions: Defined for smooth and efficient operation. Illustrate the Control Data Flow Graph (CDFG) for the system	10
2.	Design and analyze a Real-Time Weather Forecast Station that continuously monitors weather conditions and provides up-to-date forecasts to users such as meteorologists, pilots, and emergency response teams. The system may include sensors for measuring temperature, humidity, barometric pressure, wind speed, and precipitation. Given Criteria: - Types of Events - Functional Correctness - Timeliness Evaluate the system design based on these criteria to ensure it meets the requirements for real-time weather monitoring and forecasting.	10
3.	Analyze the requirements and design challenges involved in code optimization for embedded system design. Based on your analysis, apply some suitable optimization techniques to the provided code snippet. Reorganize the code snippet to achieve optimize performance while ensuring that relevant validations are properly incorporated. <pre>int main() { float a, b, c, r1, r2, d, f, m; printf("Enter the values of a b c: "); scanf("%f %f %f", &a, &b, &c); d = b * b - 4 * a * c; if (d > 0) { r1 = -b + sqrt(d) / (2 * a);</pre>	10



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r2 = -b - sqrt(d) / (2 * a);
printf("The real roots = %f %f", r1, r2);
} else if (d == 0) {
r1 = -b / (2 * a);
r2 = -b / (2 * a);
printf("Roots are equal = %f %f", r1, r2);
} else
printf("Roots are imaginary");
f=m*m*m;
return 0;

```

4. a) For the given dataset, check that RMS creates a feasible schedule. If the current execution times do not allow for a feasible schedule, adjust the execution times, create new values, and try RMS scheduling again. Draw a task timeline graph for at least three cycles. The dataset is as follows:

Task	Execution Time	Period
T1	5 <u>2</u>	15
T2	7 <u>3</u>	10
T3	10 <u>4</u>	20

b) Evaluate how do the following parameters affect the schedulability of real time tasks:-

- Arrival Time
- Current Time / Scheduling point
- Execution Time
- Rate or Period
- Deadline

Suggest an optimal scheduling scheme using any three relevant parameters as mentioned above and apply it for the following dataset:- Task Arrival Time Execution Time Period Deadline

Task	Arrival Time	Execution Time	Period	Deadline
T1	0	5 <u>2</u>	20	6
T2	0	4 <u>2</u>	15	3
T3	0	4 <u>4</u>	10	9

Note: Consider current time as per the scheduling points. Illustrate the task time-line graph for at least three cycles.

5. A database system allows multiple clients to read data simultaneously, but only one client can write data at a time. Using POSIX semaphores, implement a solution to:

- Allow concurrent reads.
- Ensure exclusive access for writes.
- Prevent starvation of readers or writers

Explain the role of each semaphore and provide the implementation with suitable code.

$$U \leq n$$

$$U = \text{Total CP}$$

$$n = \text{No. of Tasks}$$

$$U = 1.5$$

10

RMS

1) States: Idle - waiting for tasks
 Move - Forward
 Move - Backward
 Turn left
 Turn right
 Pick
 Set, place, halt

Events:
 Pick command
 If detected
 Sort - based on attributes
 Place: bin
 Bin empty
 Bin full - true & false

Idle - Pick Move
 Give
 MF - O.T - TL/TR
 MF - O.T - Pick
 Place - Set - Set
 Set - Page 2 of 2
 Place - BE - MB
 MF - OT - TL/TR
 OF - 100%
 100%